

Date: _____

Day: **M T W T F S**Def #01:-

A function of two variables, f , is a rule that assigns each ordered pair of real numbers (x, y) in a ~~$\mathbb{R}^2$~~ , $D \subset \mathbb{R}^2$, a unique real number denoted by $f(x, y)$. The set D is called the domain of f and its range is the set of values that f takes on that is $\{f(x, y) : (x, y) \in D\}$.

Domain = $(x, y) \rightarrow$ Input / Independent

Range = (z)

$\mathbb{R}^2$ = Set of real numbers in 2D (x, y)

Definition:-

Let f be a fn: of two variables, and $z = f(x, y)$. Then x and y are called independent variable and z is called dependent variable.

Domain & Range:-

All Real numbers except (2)

Ex:- Let $f(x, y) = \frac{\sqrt{x+y+1}}{x-1}$, Evaluate the $f(3, 2)$

and gives its domain.

Solution:-

$$f(x, y) = \frac{\sqrt{x+y+1}}{x-1}$$

$$f(3, 2) = \frac{\sqrt{3+2+1}}{3-1}$$

$$= \frac{\sqrt{6}}{2} \rightarrow \text{Range.}$$

in under-radical the value should be positive or in a quotient form $\frac{p}{q}$ q should be undefined.

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Domain:-

$$(x + y + 1) \geq 0 \text{ and } x \neq 1.$$

If we put $x = 1$, then function will be discontinuous.

$$\boxed{y \geq -x - 1} \rightarrow \text{inequality}$$

$$y = -x - 1 \rightarrow \text{function}$$

$>$ $\rightarrow$ upward graph

$<$ $\rightarrow$ downward graph.

Graph:-

Domain = function which values gain.
Range = Domain ki corresponding me image milegi.

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Day: MTWTFS

Exp:- Find domain of $f(x, y) = x \ln(y^2 - x)$.

Solution:-

$$y^2 - x > 0$$

$$-x > 0 - y^2$$

$$-x > -y^2$$

Domain $\left[x < y^2 \right] \rightarrow \text{inequality.}$

$x = y^2 \rightarrow \text{function.}$

Parabolic eq.

Exp:- Find the domain of $f(x, y) = \sqrt{9 - x^2 - y^2}$

Solution:-

radical is positive always.

$$9 - x^2 - y^2 \geq 0$$

or

$$-x^2 - y^2 \geq -9$$

$$-(x^2 + y^2) \geq -9$$

$$x^2 + y^2 \leq 9$$

$$D = \{ (x, y) \mid x^2 + y^2 \leq 9 \} \quad \text{line} = x^2 + y^2 = 9$$

~~Part~~ Graph:- Circle

Range:- $[0, 3]$ or

$$[0 \leq x, y \leq 3]$$

$$= \sqrt{9 - x^2 - y^2}$$

$$= \sqrt{9 - (-3)^2 - 0}$$

$$= \sqrt{9 - 9 - 0}$$

$$= \sqrt{0}$$

↓ defined